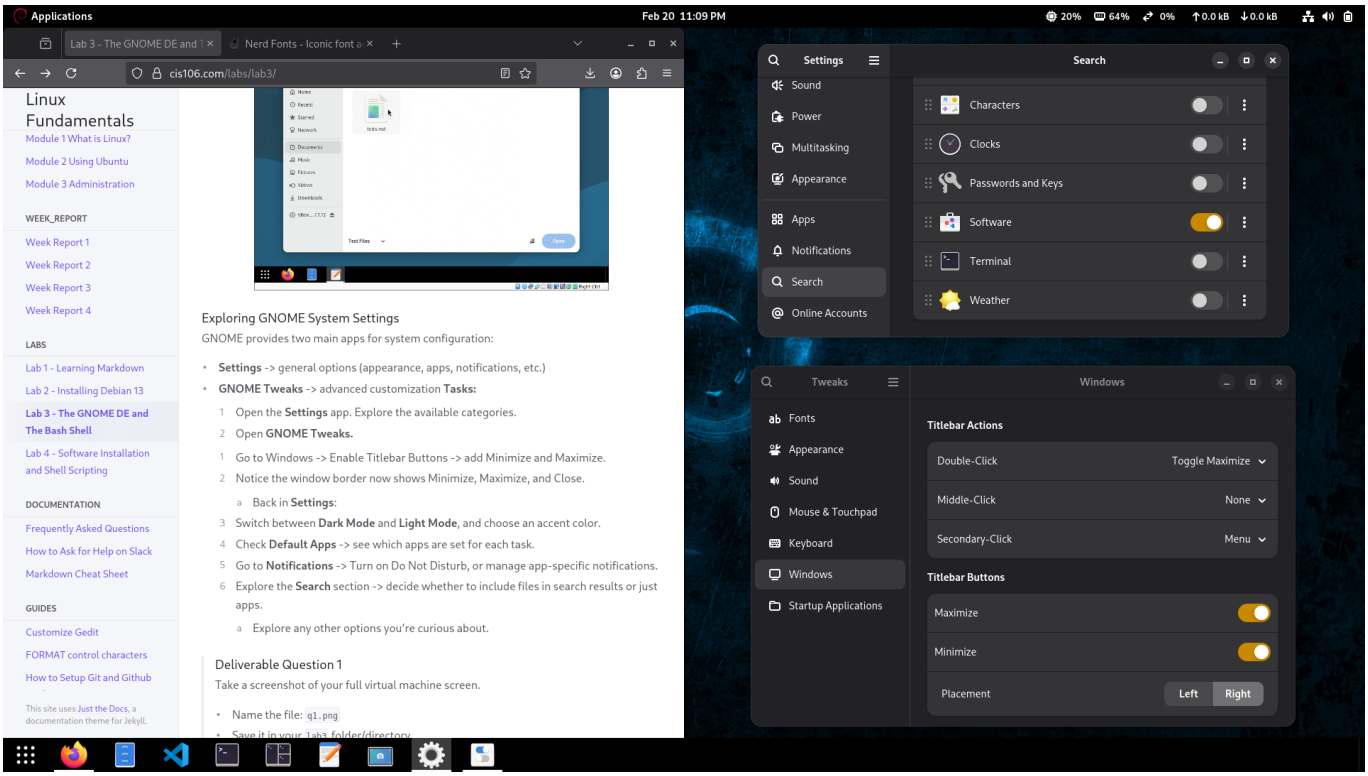
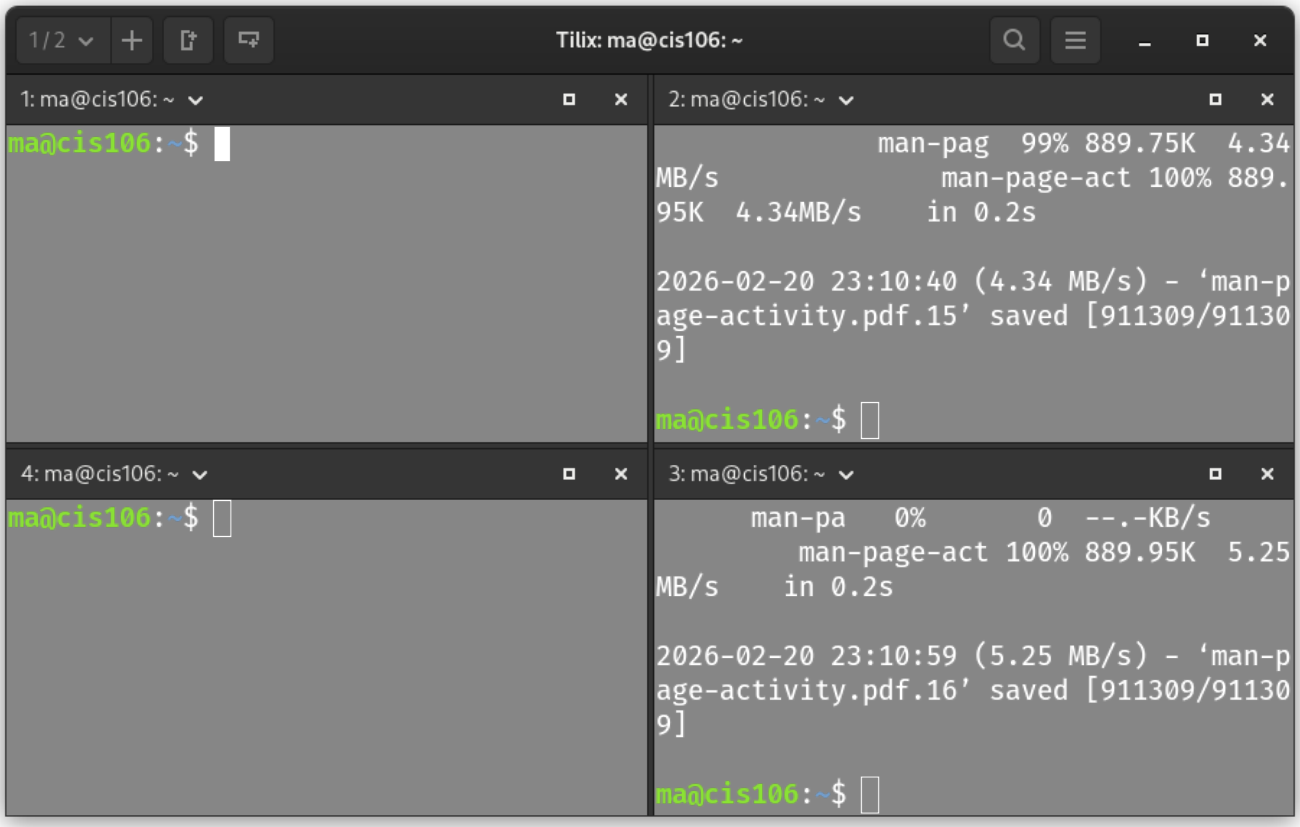
Lab 3 Submission

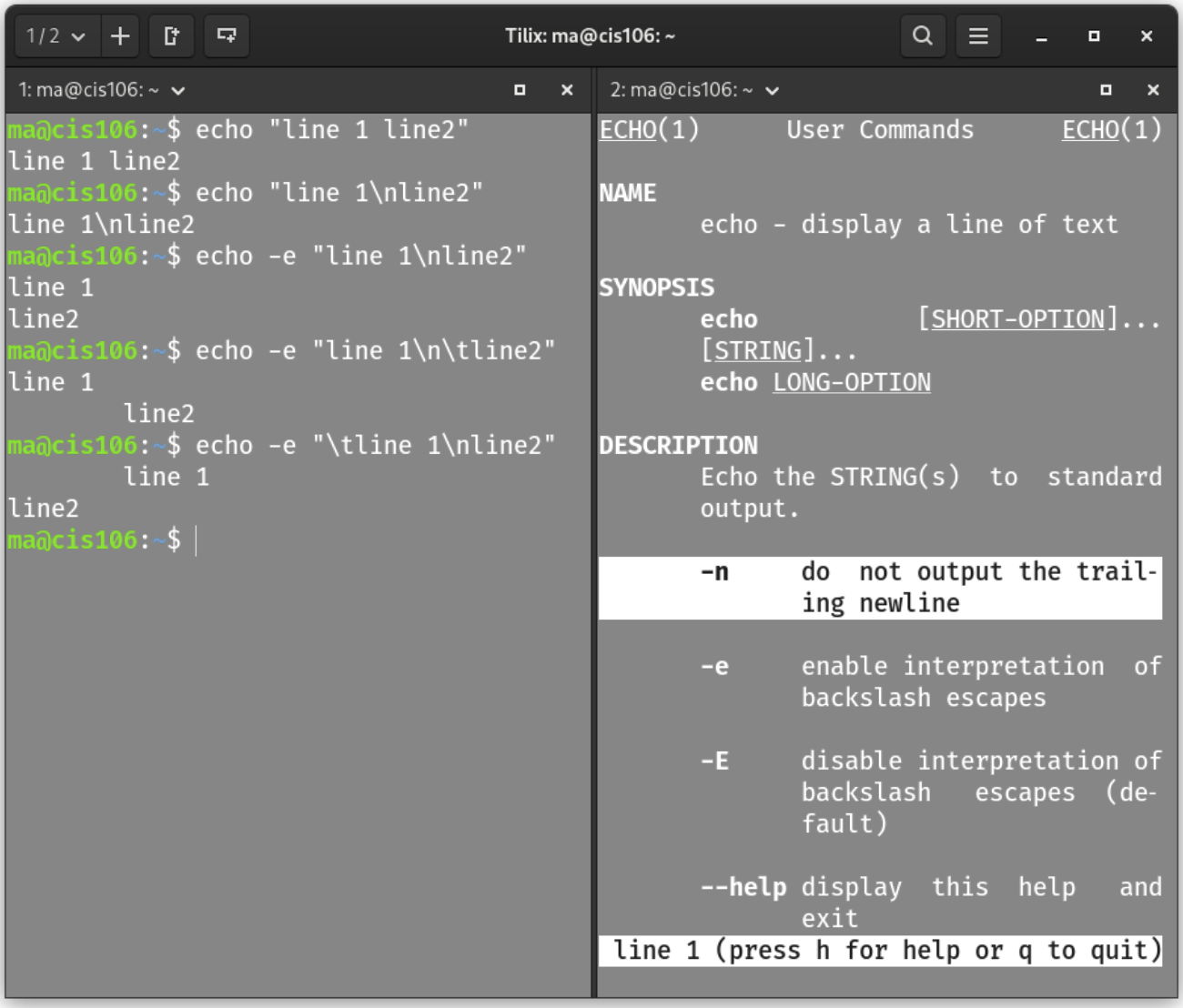
Question 1



Question 2



Question 3



Challenge Question

```
1/1                                Tilix: ma@cis106: ~
1: ma@cis106: ~                    2: ma@cis106: ~
ytes
     --peta          show output in petab
ytes
 -k, --kibi          show output in kibib
ytes
 -m, --mebi          show output in mebib
ytes
 -g, --gibi          show output in gibib
ytes
     --tebi          show output in tebib
ytes
     --pebi          show output in pebib
ytes
 -h, --human         show human-readable
output
     --si            use powers of 1000 n
ot 1024
 -l, --lohi          show detailed low an
d high memory statistics
 -L, --line          show output on a sin
gle line
 -t, --total         show total for RAM +
 swap
 -v, --committed     show committed memor
y and commit limit
 -s N, --seconds N   repeat printing ever
y N seconds
 -c N, --count N     repeat printing N ti
mes, then exit
 -w, --wide          wide output

     --help     display this help and exi
t
 -V, --version  output version informatio
n and exit

For more details see free(1).
ma@cis106:~$
```

```
1/1                                Tilix: ma@cis106: ~
1: ma@cis106: ~                    2: ma@cis106: ~
              have  exbibyte of RAM or swap,
              the number is in tebibytes and
              columns might not  be  aligned
              with header.

       -w, --wide
              Switch  to  the wide mode. The
              wide   mode   produces   lines
              longer  than 80 characters. In
              this mode  buffers  and  cache
              are  reported  in two separate
              columns.

       -c, --count count
              Display   the   result   count
              times.   Requires  the  -s op-
              tion.

       -l, --lohi
              Show  detailed  low  and  high
              memory statistics.

       -L, --line
              Show  output on a single line,
              often used with the -s  option
              to  show memory statistics re-
              peatedly.

       -s, --seconds delay
              Continuously display  the  re-
              sult   delay   seconds  apart.
              You may actually  specify  any
              floating  point number for de-
              lay using either .  or  ,  for
              decimal  point.   usleep(3) is
              used for  microsecond  resolu-
              tion delay times.
e(1) line 117 (press h for help or q to quit)
```

```
1/1                                Tilix: ma@cis106: ~
1: ma@cis106: ~                    2: ma@cis106: ~
ma@cis106:~$ uname
Linux
ma@cis106:~$ uname -s -r -v -o
Linux 6.12.73+deb13-amd64 #1 SMP PREEMPT_
DYNAMIC Debian 6.12.73-1 (2026-02-17) GNU
/Linux
ma@cis106:~$
```

```
DESCRIPTION
       Print   certain  system  information.
       With no OPTION, same as -s.

       -a, --all
              print all information, in  the
              following  order,  except omit
              -p and -i if unknown:

       -s, --kernel-name
              print the kernel name

       -n, --nodename
              print the network  node  host-
```

```
              name

       -r, --kernel-release
              print the kernel release

       -v, --kernel-version
              print the kernel version

       -m, --machine
              print   the  machine  hardware
              name

       -p, --processor
              print   the   processor   type
              (non-portable)

       -i, --hardware-platform
              print  the  hardware  platform
              (non-portable)

       -o, --operating-system
              print the operating system
ame(1) line 8 (press h for help or q to quit)
```

```
1/1                                   Tilix: ma@cis106: ~
1: ma@cis106: ~                       2: ma@cis106: ~
ma@cis106:~$ date --rfc-3339=ns
2026-02-20 23:49:13.682285356-05:00
ma@cis106:~$
```

```
       -I[FMT], --iso-8601[=FMT]
              output date/time in  ISO  8601
              format.   FMT='date'  for date
              only (the  default),  'hours',
              'minutes',  'seconds', or 'ns'
              for date and time to the indi-
              cated   precision.    Example:
              2006-08-14T02:34:56-06:00

       --resolution
              output  the  available resolu-
              tion  of  timestamps  Example:
              0.000000001

       -R, --rfc-email
              output  date  and  time in RFC
              5322 format.  Example: Mon, 14
              Aug 2006 02:34:56 -0600

       --rfc-3339=FMT
              output date/time in  RFC  3339
              format.    FMT='date',   'sec-
              onds', or 'ns'  for  date  and
              time  to  the indicated preci-
              sion.    Example:   2006-08-14
              02:34:56-06:00

       -r, --reference=FILE
              display  the last modification
              time of FILE

       -s, --set=STRING
              set time described by STRING

       -u, --utc, --universal
              print or set Coordinated  Uni-
te(1) line 33 (press h for help or q to quit)
```